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Lawal, Kelvin N.; Olaniyi, Titus K.; Gibson, Ryan M.

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A Fog Computing Based Efficient Data Management Smart Home Architecture

Kelvin N. Lawal and Titus K. Olaniyi and Ryan M. Gibson

School of Computing, Engineering and Built Environment
Glasgow Caledonian University, United Kingdom

Abstract. Smart home technology comprises of multiple devices communicating with each other and human occupants to provide state of the art living conditions. The increased use of smart devices in smart homes, increases the rate of energy consumption. Among the plethora of smart devices within a smart home, one of the key smart devices that account for high data volume and rate of data transmission is the Closed-Circuit Television (CCTV) surveillance system. Furthermore, there has been multiple attempts by researchers to use Cloud computing solution to address the issue of storage and processing capabilities. However, the use of Cloud computing has challenges such as energy consumption, latency, and network bandwidth usage bottlenecks with energy cost set to increase by April 2022. The introduction of Fog computing as a solution aim to address existing Cloud computing issues such as energy consumption, latency, and network bandwidth usage, providing improvements to mobility, security and on demand requests. In this paper, Fog computing will provide a method that supports the management of data generated by CCTV camera system while reducing the energy consumption, latency, and network bandwidth usage. The work presented in this paper demonstrates efficiency and optimisation of energy consumption, latency, and network bandwidth usage using iFogSim2 network simulation toolkit. A comparison and interpretation of results from the iFogSim2 output for Cloud and Fog based scenarios is used to evaluate improvements in energy consumption, latency, and network bandwidth usage demonstrating the benefits of Fog based scenario to efficiently manage data from the CCTV camera system.

Keywords: Fog Computing, Cloud Computing, Data Management, Internet of Things, Smart Home.

1 Introduction

Over the last decade, adoption, and advancements of the Internet of Things (IoT) technology have been growing and gaining traction at a staggering rate, where governments, organisations, households, and individuals are now adopting and implementing next generation digital system technologies, devices and components into their respective environments, business operations and homes to remain relevant and up to date in the face of changing global technological environments [1]. Furthermore, individuals have gravitated towards living in modern and urban homes interconnected with multiple IoT devices that perform various automated functions that provide support, comfort, security and easier living within the home, this home is called a smart home [2]. As part of their core function and reaction to dynamic scenarios within the smart home, these interconnected devices which are facilitated by the IoT technology regularly and intermittently communicate and transfer data to other devices, humans, and manufacturers seamlessly with minimal interference [3]. In a smart home today, heterogeneous devices are interconnected within an IoT network to generate, collect, process and transfer data through sensors and actuators to better serve the inhabitants and make living a more efficient and interactive experience. With substantial improvements in technological developments, IoT has evolved and developed simultaneously which has led to the growth of various smart (IoT) devices across the economic spectrum consequently, providing benefits such energy efficiency, convenience, and automated functions [4]. In addition, the use of Artificial Intelligence (AI) or smart applications with Closed Circuit Television (CCTV) system is used to monitor and provide security to smart homes at the basic level. However, CCTV cameras have also evolved and advanced technologically now providing detection capability through optical sensor technology embedded within the smart optical cameras.

These smart cameras can also be automated and used in law enforcement for facial recognition, manufacturing industries for product defect detection, health and safety monitoring in construction sites, transportation with smart traffic lights and motorways, in the health care for patient monitoring and in smart homes for security and intruder

detection. In smart homes, security and wellbeing are mandatory in preventing unwanted intrusion, accidents, and unwanted events. Traditionally, CCTV cameras placed at strategic locations are used in homes to address these concerns and provide the monitoring, assessment and analysis required. However, traditional CCTV cameras capture images while streaming continuously and do not provide warning, alert, or notification when there is suspicious activity [5]. Furthermore, CCTV system is physically connected, and the data is stored onsite which has its own drawbacks and security risks associated with the design. Some of these drawbacks include cost of onsite storage, servers, and staff to monitor hundreds of cameras onsite. Furthermore, the associated security risk of theft and fire onsite which could lead to loss and damage of CCTV video footage recordings. To address the issue of cost, the adoption of an IoT node system comprised of distributed video cameras could provide online streaming capability [6]. Furthermore, Cloud based computing was introduced to shift the burden from onsite to the Cloud providing processing, computation, and storage capabilities with service response in real time along with wireless smart cameras to improve and eliminate these associated security risks. In addition, modern digital and optical CCTV cameras embedded with motion detector sensors and Passive Infrared (PIR) motion sensors amongst others, are now capable of providing more advanced security and monitoring systems. These modern advanced security and monitoring systems interact with humans and other machines to provide a more secure home, building (internal) or external environment in real time.

There are major challenges in the data management process as the smart CCTV cameras generate a large amount of data consistently and overall. Within the Cloud based computing design, data needs to be collected, processed, analysed, transferred, and stored consequently, the data management process results in high energy consumption, severe bottlenecks within the network bandwidth usage and subsequently, increased latency [5]. Fog computing layer exists between the Cloud computing layer and the device layer comprising of various resources such as access points, routers, switches, and servers. Furthermore, these Fog computing resources have limited capacity and are heterogeneous regarding transient storage, processing, and computational capabilities. Consequently, the management of these Fog computing resources becomes a priority to ensure Quality of Service (QoS) through effective resource allocation [7], [8].

[9] indicates the challenges associated with the use of Cloud based computing which include higher cost of communication, increased latency, higher energy consumption and severe bottlenecks in the network bandwidth usage. Although, the Cloud possesses robust computing, processing, and storage capabilities, with most applications hosted in the Cloud, the share distance between the smart devices and the Cloud contributes to the some of the challenges associated with its use. Consequently, there has been a shift with researchers proposing the use of Fog computing as an extension to the Cloud [8]. CCTV monitoring and surveillance solutions especially within a smart home are not only relevant but critical to safety and security as modern cameras possess optical sensors for object and facial recognition technology [10], [11].

This paper will focus on the management of CCTV data within a smart home highlighting the challenges associated with the use of Cloud based computing design scenario and how this can be improved with the use of data warehouse technology and in addition, introduce Fog based computing to significantly optimise the improvements. As a result, to mitigate, improve and optimise these challenges, this paper proposes a Fog based smart home CCTV data management system with the following contributions.

- The optimisation of a smart home Cloud based network architecture using Fog computing for the management of CCTV camera data. The use of Fog nodes on each floor of the smart home to process captured CCTV recording on a monitoring screen at the security office. This could alert homeowners, security staff or law enforcement in real time to act should there be suspicious activity detected.
- A further contribution is about the issue of increased energy consumption, latency, and network bandwidth usage, using the Fog based scenario to manage resources and efficiency. The Fog nodes provide computation process and transient storage closer to the networks edge consequently, reducing energy consumption, latency and network bandwidth usage which saves cost on a micro scale and contributes to the United Nations (UN) Sustainable Development Goals (SDGs) and the overall drive towards net zero on a macro scale.
- The final contribution will demonstrate the use of iFogSim2 network simulation software toolkit to model and build a smart home network topology that will effectively evaluate the proposed Fog based smart home CCTV data management system. This contribution will further demonstrate the impact a Fog based network architecture will have on the smart home network regarding energy consumption, latency, and network bandwidth usage.

As shown in fig 1, a smart home network architecture typically comprises of multiple types of sensors and smart devices that obtain information through the sensors when there is change to the environment. This information is captured as data, processed, and analysed with a feedback contrivance for notification through alert in real time. These sensors play a key role and vary across multiple functions in a smart home such as object recognition, motion detection sensors to trigger smart lighting, CCTV smart cameras, smart meters cooling and heating control, energy management controllers, etc [3], [5], [12].

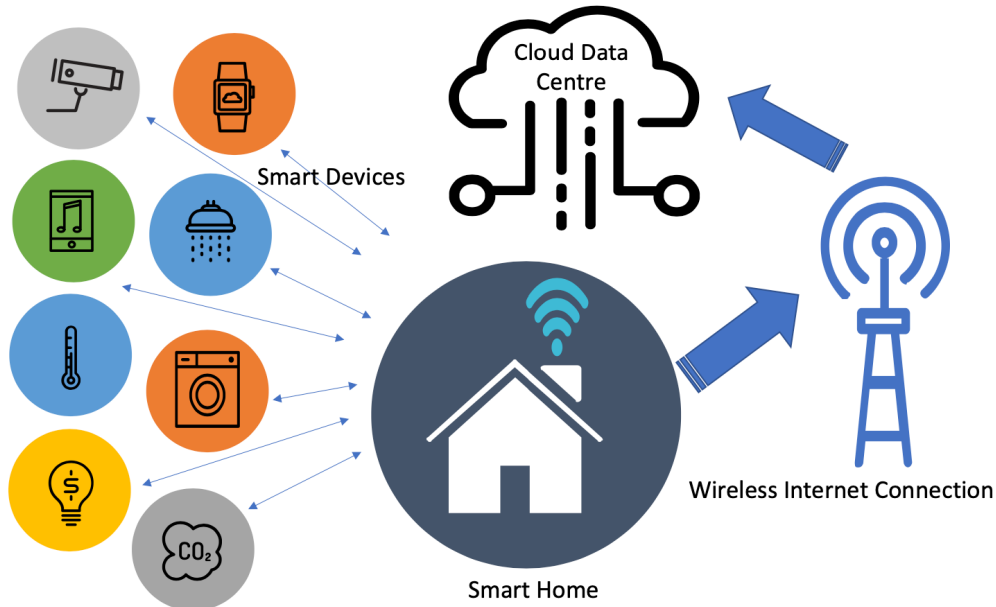


Fig. 1. A typical smart home architecture with interconnected IoT sensors and devices

2 Case Study

This section will use the smart home floor plan of a modern urban UK building to highlight the smart optical CCTV camera positions and other sensors such as CO₂ detection sensors, smart lighting sensors and infrared door sensors. Although there are multiple sensors and devices generating data within the smart home, the focus of this paper will be the smart optical CCTV camera system data management process. As shown in fig 2, the smart home floor plan comprises of 3 floors, ground floor, first floor and the second floor. On the ground floor, the floor plan shows 4 smart optical CCTV cameras identified as Data Source Silos (DSS). while on the first and second floors there are 3 DSS smart optical cameras respectively. The cameras in the smart home are strategically positioned to capture images of anyone entering or exiting the building and on each floor.

As a result, these captured images are used to monitor movement and other activities on each floor prompting real time response to investigate unusual and suspicious behaviour in the building. The average size of a captured CCTV video file is quite large and due to the size, the cameras are programmed to only activate once the embedded infrared sensor detects movement within the coverage area. This data is then sent to the Fog nodes for processing and storage while the video image output is sent to the monitoring screens in security office or to smart phones for mobile monitoring capability.

Furthermore, the smart optical CCTV cameras are strategically positioned for extensive area coverage on every floor within the smart home with microprocessors acting as a conduit between the smart optical CCTV cameras and the Fog nodes on each floor which subsequently transfers data to the Cloud layer when required. The Fog nodes are all interconnected and communicate with each other, sharing information to determine availability and capacity for computation and processing. Therefore, if the Fog node on the ground floor is busy the data is off loaded to the next available Fog node on a different floor and the image is processed and updated to the monitoring screens in the security office or smartphones.

The increasing number of IoT devices, including Machine to Machine (M2M), Machine to Human (M2H) communication as well as the core data management function create challenges. Some of these challenges include high energy consumption, latency, network bandwidth usage, data storage and computational resource constraints

[4]. To address these challenges, this section will evaluate how these challenges can be mitigated through a simulation experiment exercise using a network simulation software toolkit called iFogSim2.

The iFogSim network simulation toolset is an extension of the CloudSim network simulation toolset, it is open source and assists with the simulation of Cloud and Fog based network environments [1]. The iFogSim2 network simulation software toolset is an extension to iFogSim and has been improved to address some of the challenges in the predecessor software simulation software toolset iFogSim. These improvements include supporting application migration and loose-coupling services, logical grouping of Fog computing nodes as required. In addition, as shown in fig 2, the overview clearly depicts (in the blue boxes) the inclusion of three new components to the iFogSim2 simulation capabilities they are clustering, microservices and mobility while still maintaining all the inherited core components (in the grey boxes) from iFogSim and the CloudSim 5.0. Also highlighted are the integrations and interconnections between these three new component and the inherited core components [13].

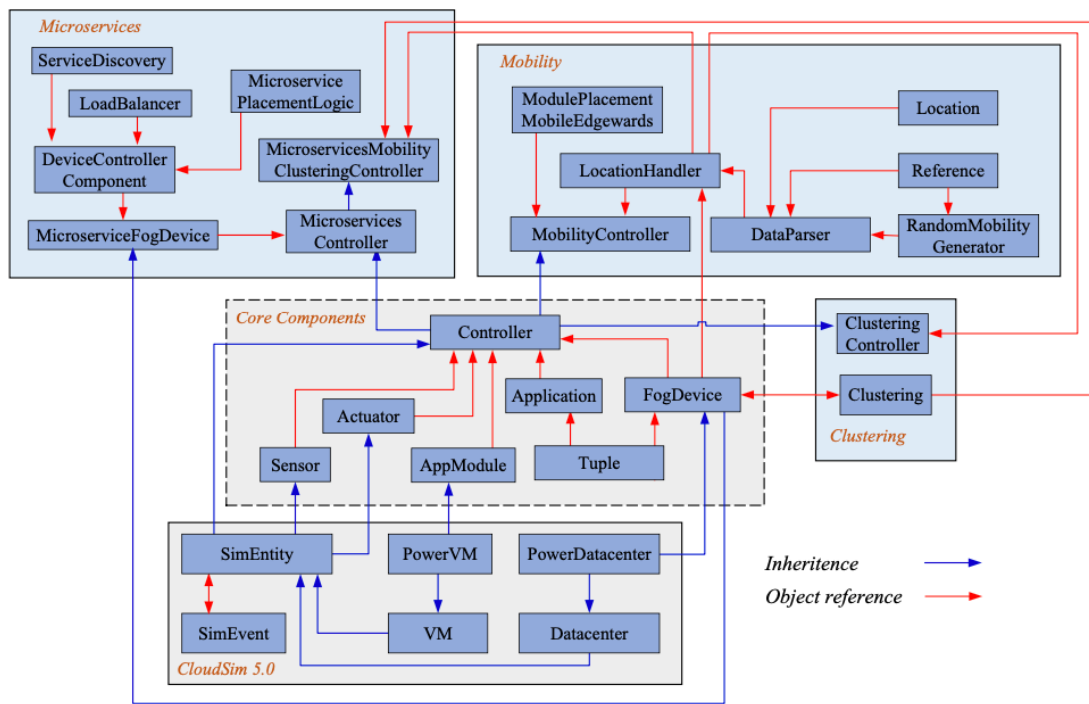


Fig 2. Overview of iFogSim2 components [13].

Furthermore, this experiment will demonstrate how the addition of the Fog computing layer will optimise the Cloud based CCTV data management scenario, providing other benefits such as mitigating the energy efficiency, latency, and network bandwidth usage challenges to the smart home CCTV data management process. Although, the standards in Fog computing have been clearly established through research, challenges still abound such as the issue of service scheduling and resource allocation limitation for data processing [11].



Fig 3. Smart home architectural floor plan

3 Proposed Solution

The proposed solution is aimed at addressing the challenges of the Cloud based computing scenario network infrastructure in the CCTV data management process such as high energy consumption, high system latency and severe network traffic bottlenecks due to high network bandwidth usage within the smart home network. The introduction of the Fog computing layer to the network architecture will improve the Cloud based computing scenario in the CCTV data management process. In the first instance, the optical CCTV cameras have embedded infrared sensors that will activate and start recording digital images when movement is detected. This recording will then be transferred to the Fog nodes which are available on each floor of the smart home. Should there be an extended amount of recording which result in the generation of large volumes of data that require processing and analysis, this will be transferred to the Cloud for processing. While Cloud computing possesses greater computing power capacity, Fog nodes are limited in their computing and processing capacity. In addition, Fog computing provides a flexible and dynamic infrastructure for optimised computing, processing, and storage capabilities therefore, reducing sever bottleneck instances in the network bandwidth usage [7].

Although the main function of CCTV cameras in a smart home is predominantly for security, CCTV provides other uses such as health care monitoring for at risk patients in a care home or hospital, defected product detection in the manufacturing industry, health, and safety compliance in the construction sector and much more. Therefore, the rationale for the adoption and use of Fog based solution in the smart home architecture is to ensure when image recordings are captured, they can be processed and analysed immediately to ensure instant notification and alert for real time reaction to potential security breaches or any risk to the smart home inhabitants. Furthermore,

Fog based solution helps reduce energy consumption, latency and network bandwidth usage with the CCTV smart optical camera data being process at the network's edge. [8].

4 Implementation

The implementation consisted of two stages; the first stage will demonstrate the Cloud based CCTV data management capability with embedded Extract Transfer and Load (ETL) process as part of the key improvement to the traditional smart home CCTV data management process. The Cloud based CCTV solution will provide centralised data reconciliation capabilities for all data generated from the smart home sensors and devices. The ETL process will ensure the data collection process is structured and cleansed allowing only essential data to be processed and managed. However, the DSS CCTV video image files are classed as big data and vary not only in volume but variety, velocity, veracity, and value [14]. According to [15], [16], on average a CCTV camera generate 25 – 30 image frames video transmission per second at the rate of 1.25Hz for a low frame rate. About 80% of CCTV cameras consume storage of up to 60GB to 600GB per camera for example, a 1080p resolution camera streaming 2MB/s will consume about 60GB. Conversely, the same 1080p camera programmed for motion-based recording will consume 600GB over the same time [17]. Consequently, the transfer of high-quality video image data from the DSS optical CCTV cameras to the Cloud over a congested or limited network bandwidth will create issues of severe bottlenecks on the network. This will subsequently increase latency especially for processing real time images and the energy consumption required to process this data will also increase.

As a result of these issues and to reduce the amount of data being transferred to the Cloud data centre, the final stage will be the introduction of the Fog computing layer, this aims to improve and optimise the Cloud based scenario when managing the DSS CCTV data from the smart home network. Additionally, Fog nodes on each floor of the smart home will perform initial analysis such as object detection, facial recognition and any additional extraction features if required where the results are forwarded to the Cloud for further analysis such as facial recognition matching, this will reduce the volume of data being transmitted over the network [11]. The output of the results can then be sent to monitoring screens in security rooms or law enforcement which can be used in real time therefore, reducing the energy consumption requirement, latency and reducing the bottleneck in the network bandwidth.

4.1 Current Posture

The Cloud based simulation scenario topology is designed and developed based on the current posture as shown in fig 4. This illustrates the Cloud data centre, connected to the smart optical CCTV cameras from DSS1 to DSS10 via the router/proxy server before being received for analysis and processing by the security and facilities department. The simulation ran directly between all available smart optical CCTV cameras in incremental order from 2 - 10 within the smart home network. This captured and measured energy consumption, latency, and network bandwidth usage of the current posture as identified in fig 4. The results obtained were used as a baseline for the evaluation of the target posture, improvements, optimisation, and management of the CCTV camera data within this smart home case study.

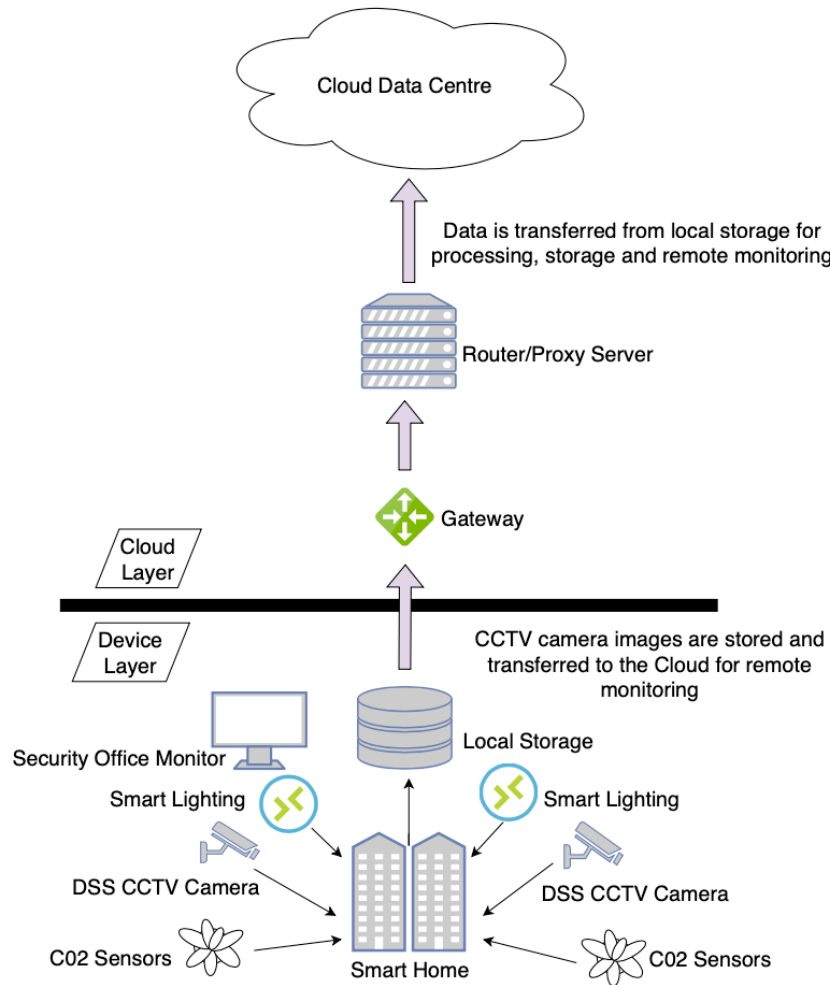


Fig 4. Cloud based CCTV data management scenario

4.2 Dataflow Management

The data management flow diagram in fig 5 clearly depicts the generation of data from the consolidated DSS CCTV cameras. The data flows to the proxy servers, the data is then passed through to the gateway where a decision is made if there is a Fog node available or not. If there is no Fog node is available, the data is then transferred to the Cloud data centre for computation, processing, and storage where the results from the data processed are then obtained by facilities and used accordingly. Should a Fog node be available, the data is transferred to the next available Fog node for computation, processing, and transient storage where Fog computing benefits apply, and the results can be obtained and used accordingly.

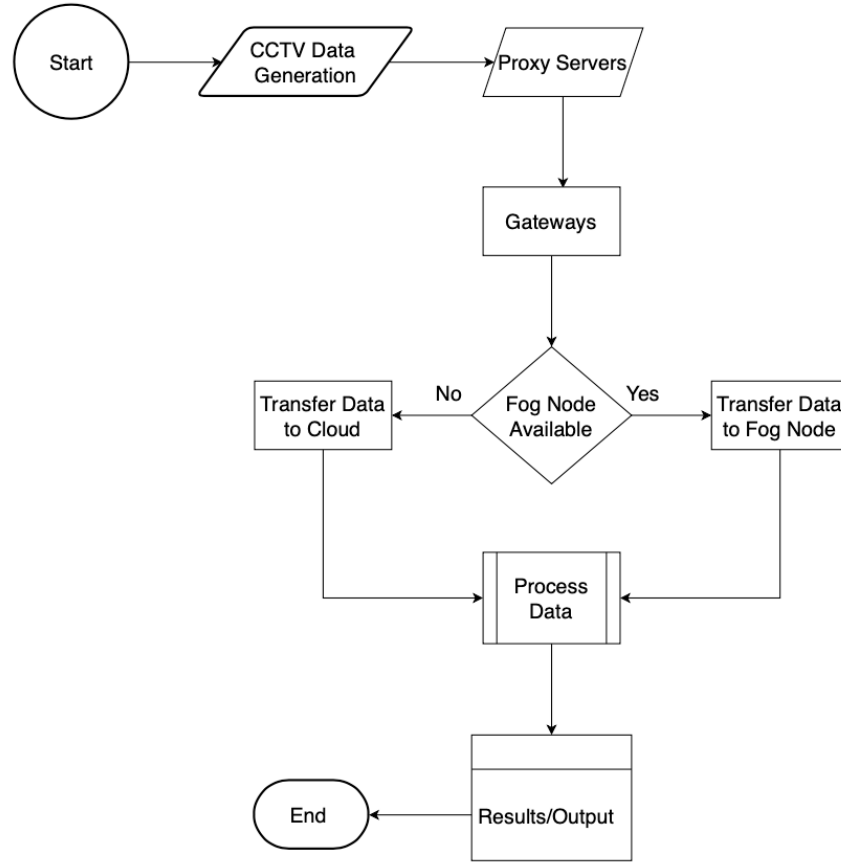


Fig 5. Data management process flow diagram

4.3 Data Management Interactions

As depicted in fig 6, this demonstrates the relationship between the DSS CCTV cameras, Cloud and Fog computing layers regarding the management of data across the entire smart home CCTV system. The Cloud computing layer plays a key role in the proposed solution for this case study, as it provides computation, processing, and storage capabilities during the data management process. The data collected from the consolidated DSS CCTV cameras is stored in the Cloud data centre at the Cloud layer where it is accessed via the proxy servers by the Fog nodes within the Fog layer upon request as required. The interaction between the Cloud data centre and the Fog node is quite simple, where the Fog nodes transfer data to the Cloud data centre when the data is no longer required through a certain time lapse period set as part of the function within the Fog nodes. When the data is required by the Fog nodes, the Cloud data centre transfers the data upon request. Fig 6 further highlights the communication bridge between the consolidated DSS CCTV cameras, Cloud data centre and the Fog nodes via the proxy server where data is generated, collected, and transferred from the DSS CCTV cameras to the Fog node where computation, processing, communication, coordination, and data synchronisation takes place during the end-to-end data management process. There is constant interoperability and communication between all Fog nodes within the Fog layer via the proxy server where Fog nodes communicate with other neighbouring Fog nodes to determine computation and storage capacity based on incoming data service requests [18]. Therefore, in this case study, the proposed solution to address the data management challenges can be optimised to further improve energy efficiency, systems latency and network bandwidth usage which subsequently improves speed and transmission of critical data for efficient and effective strategic decision making.

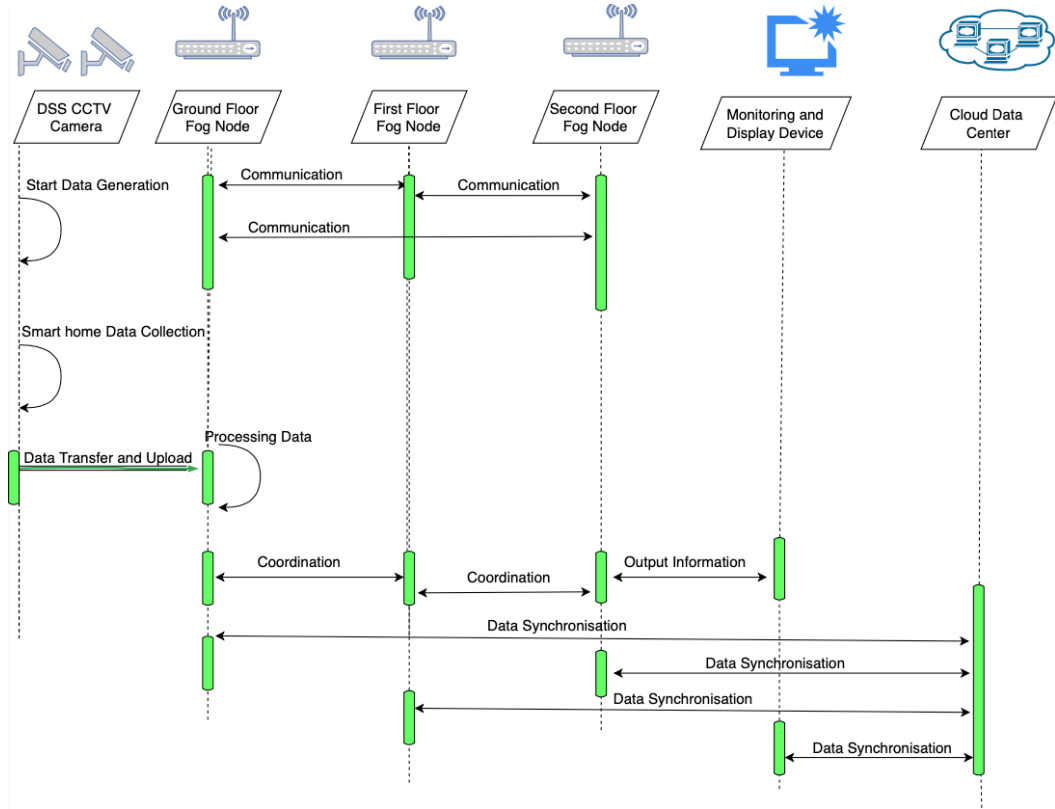


Fig 6. Data source silos, Fog nodes and Cloud interactions

5 Cloud Based Simulation Scenario

The simulation experiment comprises of DSS CCTV cameras at the device layer with one embedded sensor Device Sensor (DS) and one Device Actuator (DA) connected to each DSS CCTV camera. The sensors represent how the data is collected and the actuators are used to view and interact with the data collected from the DSS CCTV. In addition, as shown in fig 7, within the Cloud based simulation scenario illustration in iFogSim2, all the critical data is evaluated showing how data flows through to the data centre at Cloud computing layer where large and high-level computations and storage capabilities abound. The data is then transferred through the router/proxy server where the scheduling take place via the connected gateway. In addition, the data is transferred to the Cloud data centre for long term storage when the CCTV data is not required by the Fog nodes. When required, a request is made to the Cloud by the Fog nodes and the Cloud data centre services this request accordingly.

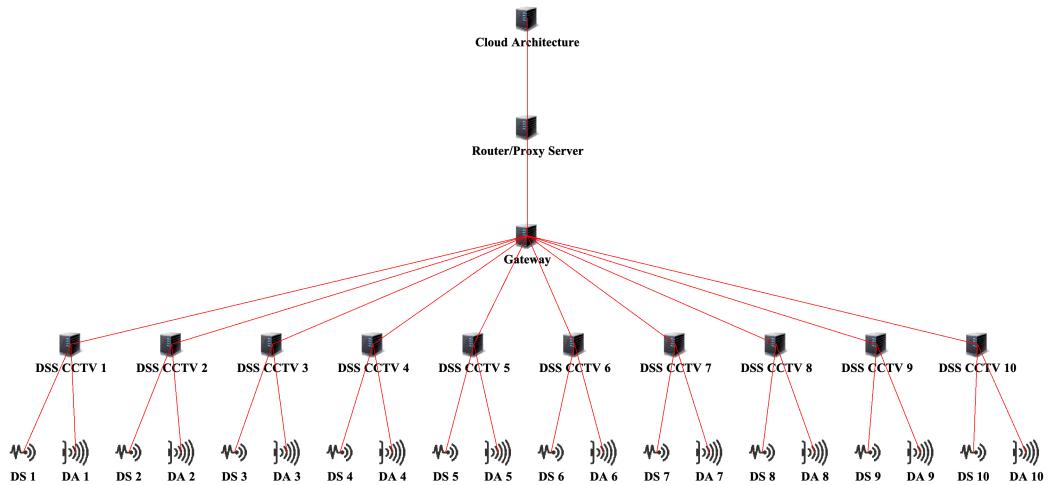


Fig 7. iFogSim2 network topology with Cloud based scenario

The information in table 1 clearly highlights the Cloud based scenario network topological configuration, design parameters and values within the iFogSim2 simulation for the Cloud, router/proxy server, gateways, and the data source silo servers respectively. As shown in table 1, these configuration parameters and values also represent bandwidth uplink and downlink, tier level, rate per MIPS, idle and busy power consumption, RAM and CPU speed MIPS for the Cloud data centre, router/proxy server, gateways, and data source silos.

Table 1. CCTV Cloud based simulation scenario parameter values.

Parameter	Cloud	Router/Proxy Server	Gateways	DSS CCTV Cameras
Tier Level	0	1	2	3
CPU Speed (MIPS)	44800	2800	5000	500
Internal RAM (GB)	40	4	4	2
Bandwidth Uplink (MBPS)	100	10000	10000	1000
Bandwidth Downlink (MBPS)	10000	10000	10000	270
Rate Per MIPS	0.01	0.0	0.0	0
Busy Power (MJ)	16*103	107.339	100.339	87.53
Idle Power (MJ)	16*83.25	83.433	80.433	82.44

While table 2 shows the application module parameter values such as the application data source, the destination of data, the CPU MIPS, tuple network parameters, the different tuple types, the data tuple direction up or down and the edge device type.

Table 2. Cloud based application parameter values

Parameter	App Module					
Source	Data from CCTV	User Module	System Processing	System Processing	User Module	
Destination	User Module	System Processing	Storage Module	User Module	Results On Screen	
Tuple CPU MIPS	2000	3500	1000	14	1000	
Tuple Network	500	500	1000	500	500	
Tuple Type	Data Sources	Raw Data	Processed Data	Action Command	Actuation Signal	
Tuple Direction	Up	Up	Up	Down	Down	
Edge Type	Sensor	Module	Module	Module	Actuator	

Furthermore, table 3 shows the different tuple mapping parameter values which include module name, the module selectivity and the input and output tuple types such as the source data, raw data, and the action command.

Table 3. CCTV tuple mapping parameter values

Tuple Mapping	Parameter				
Module Name	User Module	System Processing	System Processing	User Module	
Input Tuple Type	Source Data (Silo)	Raw Data	Raw Data	Action Command	
Output Tuple Type	Raw Data	Processed Data	Action Command	Actuation Signal	
Selectivity Module	Fractional Selectivity 1.0	Fractional Selectivity 1.0	Fractional Selectivity 1.0	Fractional Selectivity 1.0	

Table 4 shows the results of the Cloud based simulation scenario results extracted from the simulated output for DSS CCTV cameras 2 - 10. In this simulation scenario, the data sent from the DSS CCTV cameras were measured against energy consumption, latency, and network bandwidth usage. As shown in table 4, the key focus of this paper is to measure energy consumption, latency, and network bandwidth usage and to mitigate the challenges associated with the proposed solution therefore, the Cloud based simulation scenario captures the posture. The energy consumption illustrates simulated figures from 2 DSS CCTV cameras at 2676.74 kilo joules (KJ), latency

speed of 26.55 milli seconds (MS) and 21949 megabytes per second (MB/s) to 10 DSS CCTV cameras at 2696.72 (KJ) for energy consumption, latency at 33.30 (MS) and 77889 (MB/s) for network bandwidth usage.

Table 4. Cloud based simulation scenario results

DSS CCTV	Energy Consumption (KJ)	Latency (MS)	NW Bandwidth Usage (MB/s)
2	2676.74	26.55	21949
4	2683.08	27.31	43585
6	2691.58	29.52	64943
8	2693.66	30.44	68159
10	2696.72	33.30	77889

Fig 8 is a graphical illustration of the Cloud based simulation scenario output showing the difference in energy consumption on the vertical axis in (KJ) against the number of data source silos on the horizontal axis from 2 – 10. The output demonstrates that the more data source silos are added to the network the more energy is required to process and manage the data therefore, the graph shows a linear progression from 2 – 10 data source silos. While fig 9 highlights the system latency of the Cloud based simulation scenario depicting the increase in latency as the number of data source silos increase from 2 -10 which also shows a linear progression. This means that as the number of data source silos increase, the size of data being transferred to the Cloud layer for computation, processing and storage increases consequently, the response to requests will be slower which increases the latency times. Fig 10 also depicts results of the network bandwidth usage for the Cloud based simulation scenario. This demonstrates that the increase in the number of data source silos results in increased network activity which means higher levels of congestion as more data is transferred to the Cloud data centre for computation, processing and storage increasing the processing of tuples at the Cloud layer. Consequently, this relates to the slower response times in the system latency and increases energy consumption as more processing and storage is required as the number of data source silos continue to increase.

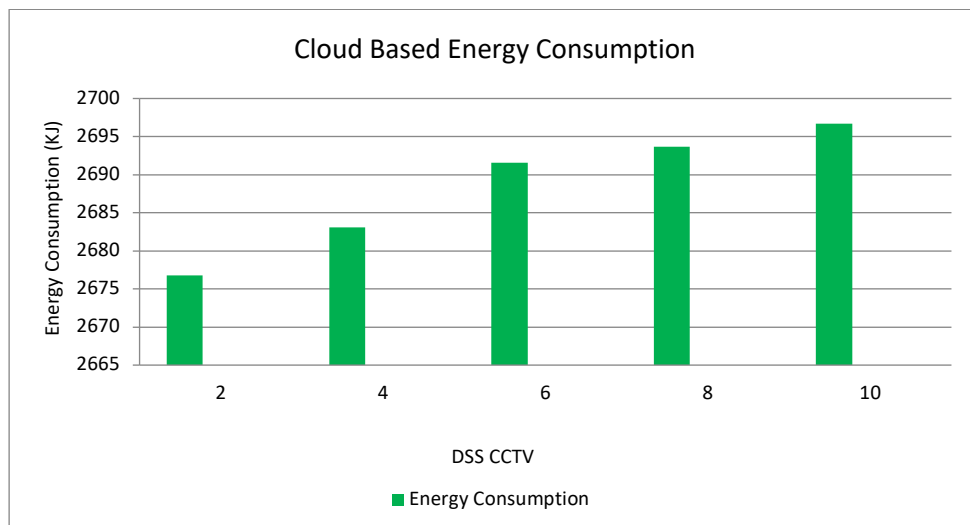


Fig 8. DSS CCTV Cloud-based simulation scenario energy consumption

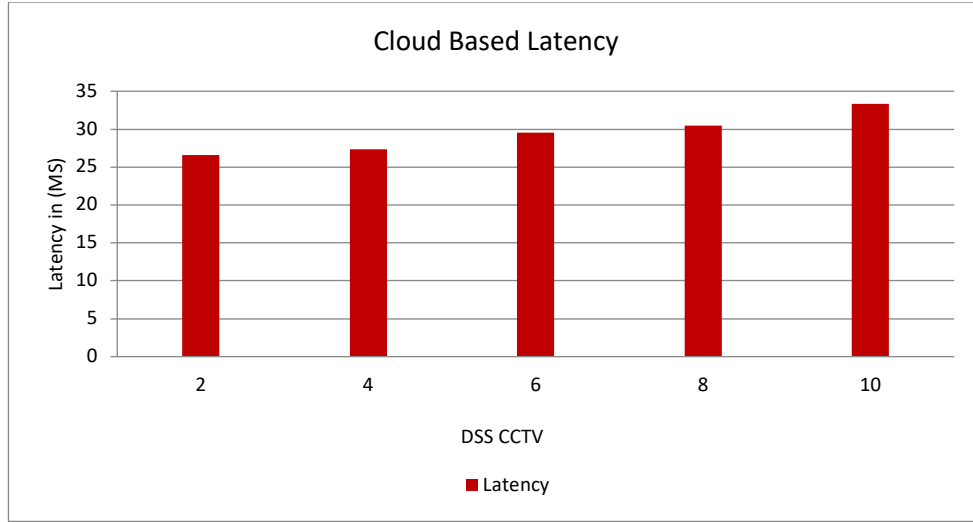


Fig 9. DSS CCTV Cloud-based simulation scenario latency

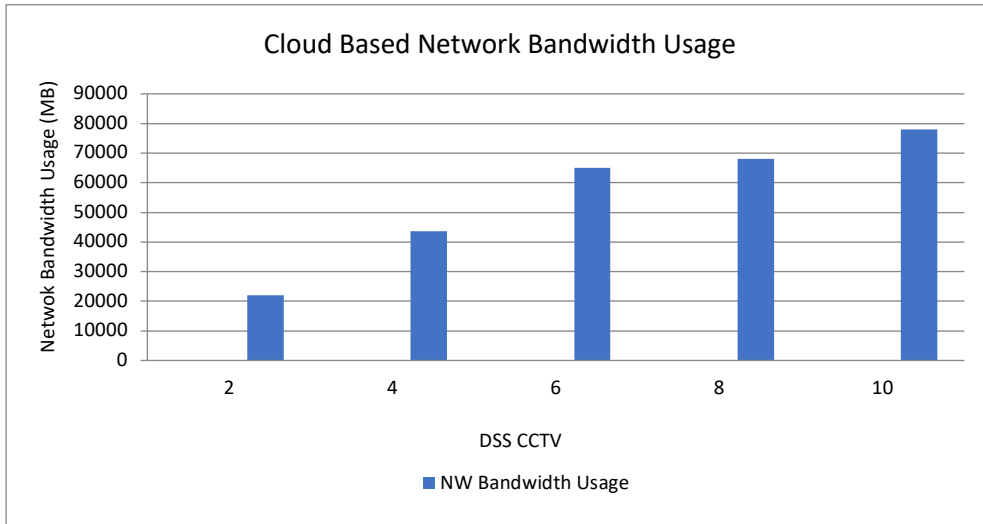


Fig 10. DSS CCTV Cloud-based simulation scenario network bandwidth usage

6 Fog Computing Optimised Scenario

The introduction of the Fog computing layer to take advantage of the benefits that Fog computing brings as an extension to Cloud computing plays an important role in the optimisation of the proposed solution to help achieve the target posture. Fig 11 clearly illustrates the introduction of the Fog computing layer to the Smart home network infrastructure. This can be summarised in similar terms to fig 4 above with the addition of the Fog computing layer. There are numerous benefits associated with Fog computing which the smart home network infrastructure aims to benefit from such as near network edge computation, processing, and transient storage. The introduction of the Fog computing layer aim to optimise the management of DSS CCTV data within the smart home network through the reduction of energy consumption, systems latency and enhancing network bandwidth usage.

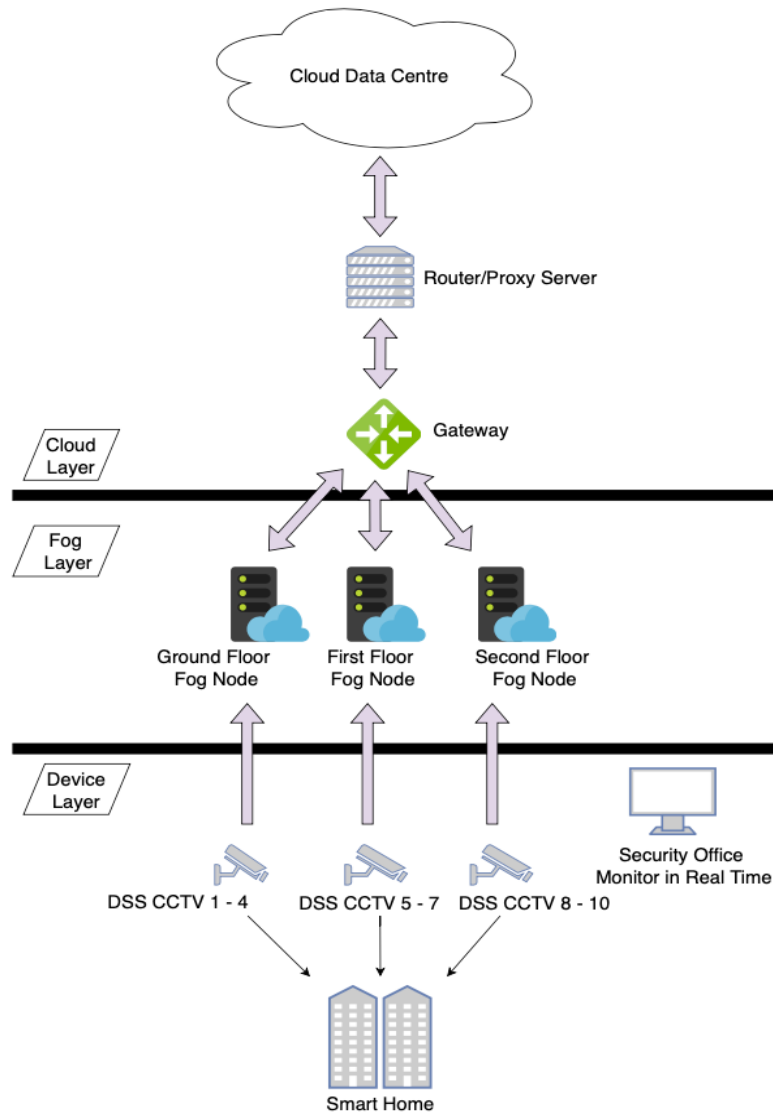


Fig 11. Proposed Fog based CCTV data management solution

6.1 Fog Computing Simulation Scenario

The Fog computing layer integration to the smart home network infrastructure comprises of extending the Cloud data centre capabilities by providing computing, processing, and transient storage at the networks edge. Consequently, this enhances the computing, data processing and storage capabilities during the DSS CCTV data management process within the smart home network. Fig. 12 illustrates the network topological diagram of the proposed solution which include the Fog computing layer represented by the Fog nodes. The Fog nodes are connected to the DSS CCTV cameras where data flows to the router/proxy server where scheduling and offloading process occurs to ensure Fog node's limited resources are efficiently used. In the iFogSim2 network simulation tool, the Fog nodes are added incrementally as required based on the service request, the data is then transferred Cloud computing layer where further improvements such as data analytics, data cleansing and AI interactions which subsequently enhances real time reactions and decision making in response to data monitoring and output.

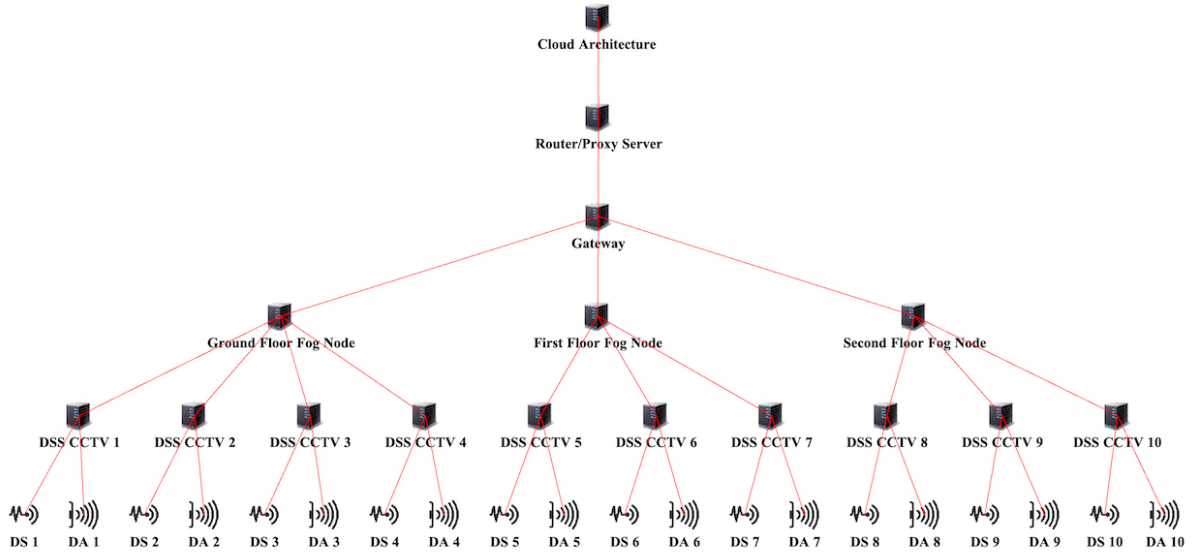


Fig 12. iFogSim2 network topology with Fog based solution

Table 5 shows parameters and values for the proposed Fog based computing scenario network topological simulation configuration design. The parameters and values represent the bandwidth uplink and downlink, tier level, rate per MIPS, idle and busy power consumption, RAM and CPU speed MIPS for Cloud data centre, router/proxy server, gateways, Fog nodes and DSS CCTV cameras.

Table 5. Fog-based solution simulation parameter values

Parameter	Cloud	Router/Proxy Server	Gateway	Fog Nodes	DSS CCTV Cameras
Tier Level	0	1	2	3	4
CPU Speed (MIPS)	44800	50000	2800	10000	1000
Internal RAM (GB)	40	80	4	4	2
Bandwidth Uplink (MBPS)	100	1000	10000	10000	1000
Bandwidth Downlink (MBPS)	10000	10000	10000	10000	270
Rate Per MIPS	0.01	0.01	0.0	0.0	0
Busy Power (MJ)	16*103		107.339	100.01	80.1
Idle Power (MJ)	16*83.25		83.433	80.1	80.06

Table 6 highlights the network simulation output results for the Fog computing layer scenario from simulating the 10 DSS CCTV cameras. The figures illustrated from the target posture simulation of 2 and 10 data DSS CCTV cameras for energy consumption are 467.07. (KJ) and 365.97 (KJ) respectively, system latency speed 9.1139999999998000 (MS) and 9.1140000000000900 (MS) respectively while network bandwidth usage is 20460 (MB/s) and 102300 (MB/s) respectively.

Table 6. Fog-based solution simulation results

DSS CCTV	Energy Consumption (KJ)	Latency (MS)	NW Bandwidth Usage (MB/s)
2	467.07	9.113	20460
4	464.16	9.113	40920
6	377.90	9.113	61380
8	367.35	9.113	81840
10	365.97	9.114	102300

Fig 13 is a graphical illustration of the Fog computing layer solution simulation output results measured in kilo joules from 2 – 10 DSS CCTV. The output results in summary shows a reverse linear progression when simulating the management of data from 2 – 10 DSS CCTV cameras within the smart home network infrastructure. Where the energy consumed for 2 DSS CCTV cameras is higher than 10. However, a linear progression applies to fig 14 for system latency and fig 15 network bandwidth usage where the management of data from 2 DSS CCTV cameras is lower than 10 DSS CCTV cameras.

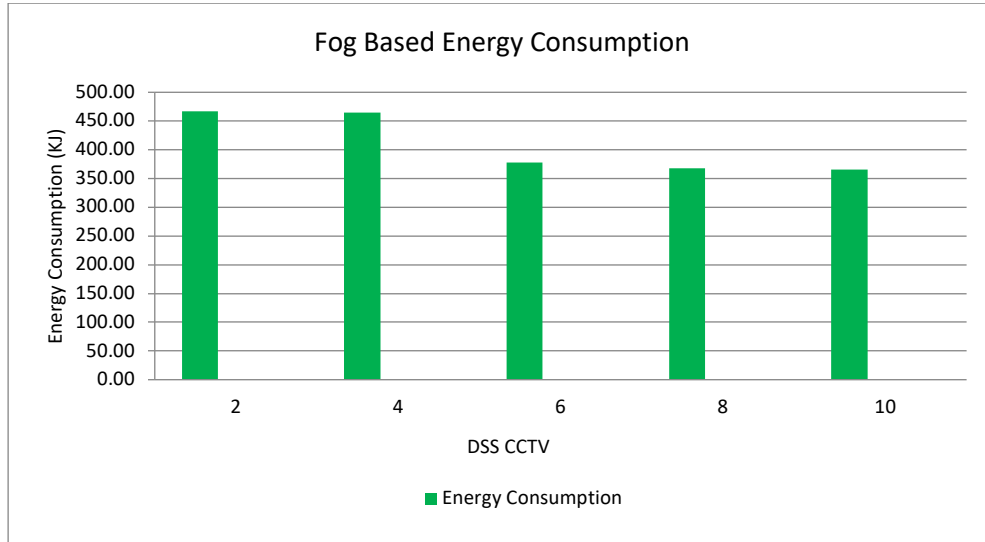


Fig 13. Fog-based solution energy consumption simulation

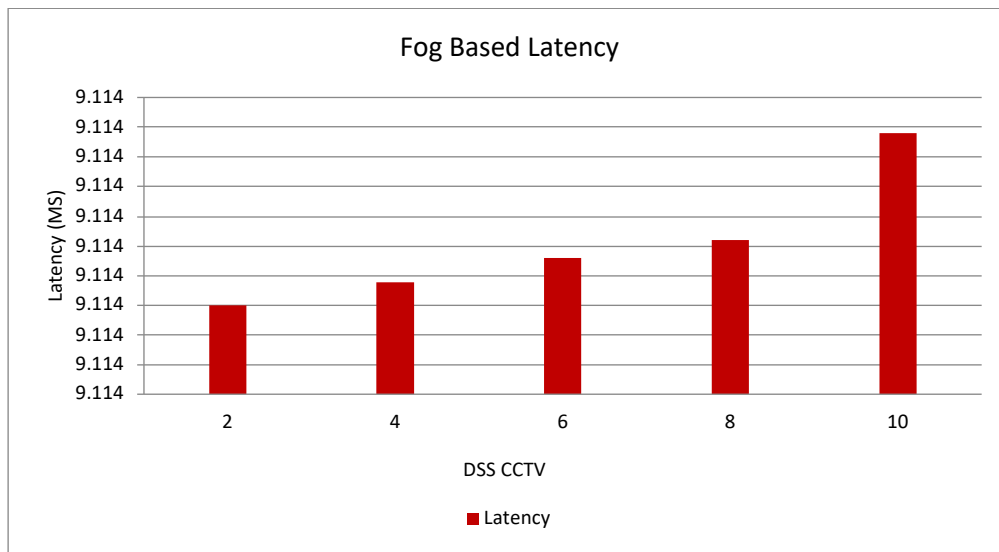


Fig 14. Fog-based solution latency simulation

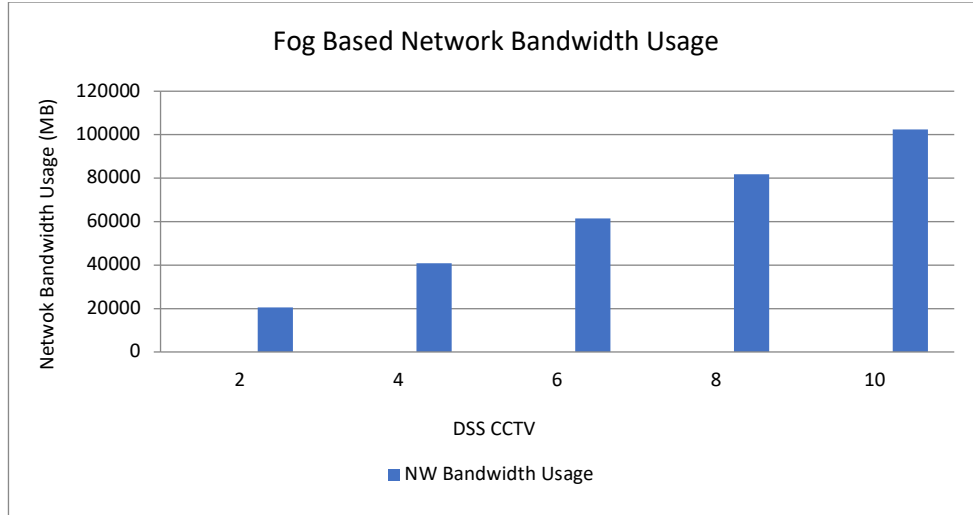


Fig 15. Fog-based solution network bandwidth usage simulation

7 Energy Consumption, Latency, and Network Bandwidth Usage Comparison

This section comprises of the comparisons between the output results from Cloud based and Fog based simulation scenarios using the iFogSim2 network simulation toolkit. The experiment was conducted with the management of data from up to 10 DSS CCTV cameras. The comparison focuses on output results for energy consumption, system latency and network bandwidth usage.

7.1 Energy Consumption Comparison Analysis

Table 7 below highlights comparison of the energy consumption simulation output results from 2 – 10 DSS CCTV using the same value and parameters to ensure the core differences across both scenarios can standout for the benefit of this simulation experiment. When simulating results from 2 DSS CCTV cameras, the Cloud based scenario output results was 2676.74 (KJ) while the Fog computing bases scenario was 467.07 (KJ). In summary for data management from 2 DSS CCTV, the Cloud based scenario energy consumption was at 54% while the Fog computing scenario was significantly lower at almost 10%. This further shows that, the use of Cloud based scenario which represents the current posture of the smart home network, there is considerable energy consumption when managing data direct from the DSS CCTV cameras. In addition, the Fog computing scenario significantly optimises the Cloud based scenario where the energy consumption level is reduced from 2420.29 (KJ) to 467 (KJ). On the other hand, the energy consumption for the Cloud based scenarios shows an increase in energy consumption as the number of DSS CCTV increase while the Fog based scenario shows a decrease in energy consumption in relation to an increase in the number of DSS CCTV cameras. This demonstrates one of the core benefits of Fog computing where computing, processing and storage of data is done at the edge of the network closer to devices rather than at the Cloud computing data centre layer. Furthermore, this highlights the efficiency of the Fog nodes, as computing and processing requests are constantly being off loaded to neighbouring Fog nodes within the smart home network infrastructure.

Table 7. Energy consumption simulation results comparison

DSS CCTV	Cloud-Based Scenario (KJ)	Fog-Based Scenario (KJ)	Percentage Difference (%)
2	2676.74	467.07	473%
4	2683.08	464.16	478%
6	2691.58	377.90	612%
8	2693.66	367.35	633%
10	2696.72	365.97	637%

As presented in fig 16, the energy consumption levels can be clearly seen across both scenarios, where the Cloud based scenario is high above the 2500 (KJ) mark however, the Fog computing-based scenario is considerably lower under the 500 (KJ) mark. Conversely, there is a reduction in the energy consumption levels as the number of DSS CCTV increase for the Fog computing-based scenario which demonstrates that the Fog computing-based scenario is more energy efficient.

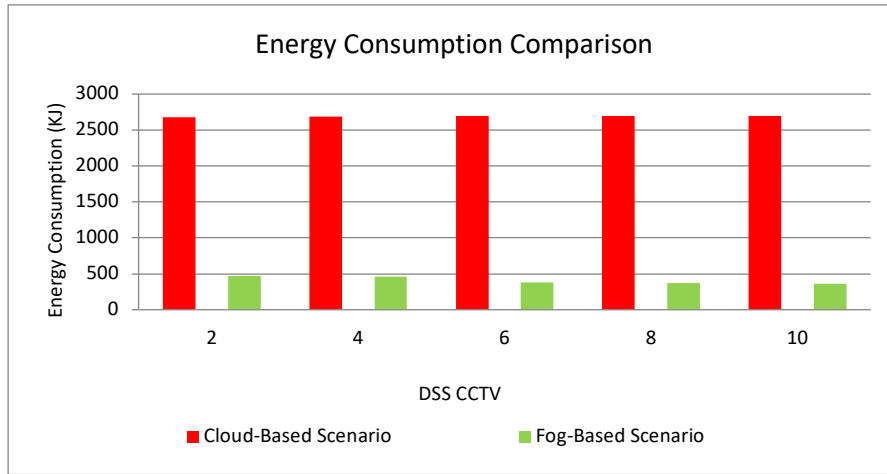


Fig 16. Energy consumption simulation comparison

7.2 Latency Comparison Analysis

As presented In table 8, the system latency simulation output results from 2 – 10 DSS CCTV cameras while maintaining the same parameters and values to clearly highlight the main differences between both scenarios. The simulation of 2 DSS CCTV between the Cloud and Fog based scenarios shows a substantial reduction in the latency level from 26.55 (MS) to 9.11 (MS). Furthermore, the differences in latency within the Fog based scenario for 2 and 10 DSS CCTV cameras is negligible at 9.1139999999998000(MS) to 9.1140000000000900(MS)

DSS CCTV	Cloud-Based Scenario (MS)	Fog-Based Scenario (MS)	Percentage Difference (%)
2	26.55	9.113	191%
4	27.31	9.113	200%
6	29.52	9.113	224%
8	30.44	9.113	234%
10	33.30	9.114	265%

As shown in fig 17, the level of system latency in milli seconds (MS) across both scenarios clearly shows an improvement in system latency through the decrease in the speed of response to service requests within the smart home network when comparing the Cloud based and Fog based scenarios.

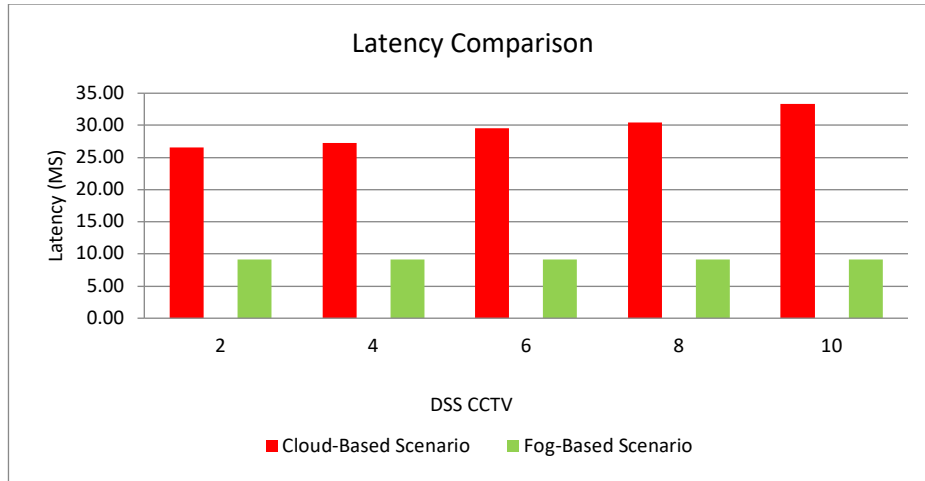


Fig 17. System latency simulation comparison

7.3 Network Bandwidth Usage Analysis

Table 9 presents the network bandwidth usage output results from 2 -10 DSS CCTV cameras. The analysis of data management from 2 DSS CCTV camera simulation highlights the clear distinction between both scenarios. Simulating the management of data from 2 DSS CCTV cameras for the Cloud based scenario is 21949 (MB) while the Fog based scenario is 20460 (MB).

Table 9. Network bandwidth usage simulation results comparison

DSS CCTV	Cloud-Based Scenario (MB/s)	Fog-Based Scenario (MB/s)	Percentage Difference (%)
2	21949	20460	7%
4	43585	40920	7%
6	64943	61380	6%
8	68159	81840	-17%
10	77889	102300	-24%

The graphical illustration in fig 18 clearly depicts the network bandwidth usage level comparison across the Cloud based and Fog based scenarios. Although, some of the differences are not that drastic such as the network bandwidth usage comparison from 2 DSS CCTV. However, as the number of DSS CCTV cameras increase the differences also increases. For instance, the increase in traffic across the smart home network during the Cloud based simulation mean that all computing, processing, and storage is done in the Cloud data centre therefore, there is an increase in network bandwidth usage.

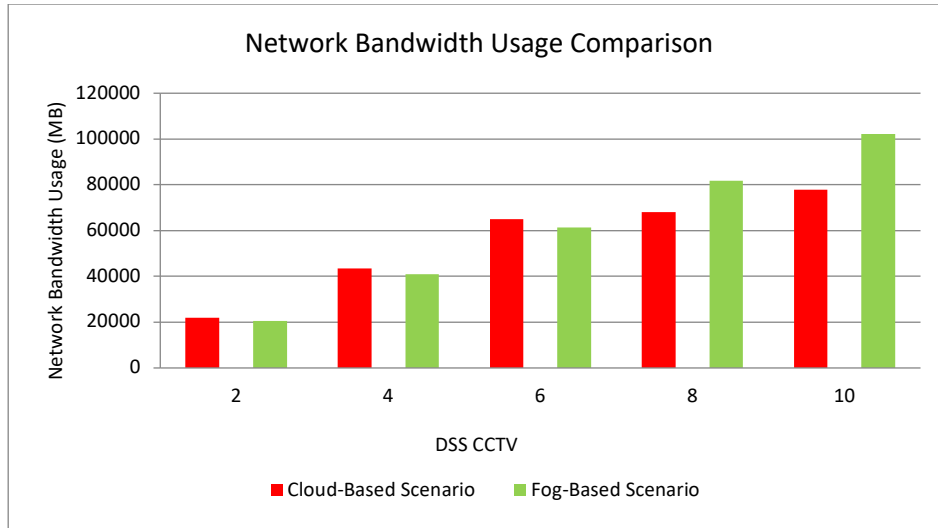


Fig 18. Network bandwidth usage simulation comparison

8 Conclusions

The management of data generated from sensors and devices such as Co2 emission, smart meter, smart white goods, wearable smart devices, smart optical CCTV cameras, etc within a smart home network infrastructure can always be improved and is an ongoing challenge. Currently, smart devices within a smart home have limited computing and processing capabilities hence the need for a centralised Cloud based data management architecture. However, Cloud based computing and data management has multiple challenges such energy consumption, latency, and network bandwidth usage. Consequently, the introduction of Fog computing addresses the challenges and provides a good level of improvement and optimisation. On the other hand, the sharp increase in smart devices over the last decade gives a clear idea on projections over the next decade. It is estimated that the number of sensors and smart devices will increase exponentially, and this will impact ability to manage data especially for applications and services that require real time response.

According to the results of the proposed Fog based solution simulation to address energy consumption, latency, and network bandwidth usage limitations within a smart home network infrastructure, it clearly shows a successful optimisation of the limitations identified. Although processing and analysis of CCTV data in real time shows an advancement in CCTV camera and sensor technology however, the issue of personal privacy is always a contentious area. In addition, the use of CCTV in a smart home further highlights privacy issues especially images from bedrooms and toilets being processed in an IoT network with data security also becoming an issue that requires further research. Additionally, there is a need to improve computing and data storage in devices and possibly sensors to help manage some aspects of data processing at the device level. This will significantly assist as the number of devices continue to increase over the next few years.

This paper highlights the setup and design configuration including the network topological design in iFogSim2. Furthermore, the output and summary results of the simulation for the Cloud based architecture and the proposed Fog based solution was evaluated and discussed. In addition, the results were used to explain how this proposed solution setup will help address issues with using Cloud based computing to manage the CCTV data. The results from the experiments further demonstrated the added benefit and optimisation of the Fog computing layer to the smart home network infrastructure.

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